~

GNU nano 8.7

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];

void* philosopher(void* num) {
    int id = *(int*)num;

    printf("Philosopher %d is thinking...\n", id);
    sleep(1);

    sem_wait(&chopstick[id]);                // Left chopstick
    sem_wait(&chopstick[(id + 1) % N]);      // Right chopstick

    printf("Philosopher %d is eating...\n", id);
    sleep(2);

    sem_post(&chopstick[id]);                // Put left
    sem_post(&chopstick[(id + 1) % N]);      // Put right

    printf("Philosopher %d finished eating.\n", id);

    return NULL;
}

int main() {
    pthread_t phil[N];
    int ids[N];

    for (int i = 0; i < N; i++) {
        sem_init(&chopstick[i], 0, 1);
    }

    for (int i = 0; i < N; i++) {
        ids[i] = i;
        pthread_create(&phil[i], NULL, philosopher, &ids[i]);
    }

    for (int i = 0; i < N; i++) {
        pthread_join(phil[i], NULL);
    }

    for (int i = 0; i < N; i++) {
        sem_destroy(&chopstick[i]);
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execu
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justi

```
hp@LAPTOP-E3MDDK1P MSYS ~
$ pacman -S gcc
warning: gcc-15.2.0-1 is up to date -- reinstalling
resolving dependencies...
looking for conflicting packages...

Packages (1) gcc-15.2.0-1

Total Installed Size:  197.99 MiB
Net Upgrade Size:        0.00 MiB

:: Proceed with installation? [Y/n]
(1/1) checking keys in keyring
(1/1) checking package integrity
(1/1) loading package files
(1/1) checking for file conflicts
(1/1) checking available disk space
:: Processing package changes...
(1/1) reinstalling gcc
:: Running post-transaction hooks...
(1/1) Updating the info directory file...

hp@LAPTOP-E3MDDK1P MSYS ~
$ nano dining_philosophers.c

hp@LAPTOP-E3MDDK1P MSYS ~
$ gcc dining_philosophers.c -o dining -lpthread

hp@LAPTOP-E3MDDK1P MSYS ~
$ ./dining
Philosopher 0 is thinking...
Philosopher 1 is thinking...
Philosopher 2 is thinking...
Philosopher 3 is thinking...
Philosopher 4 is thinking...
Philosopher 0 is eating...
Philosopher 4 is eating...
Philosopher 0 finished eating.
Philosopher 4 finished eating.
Philosopher 3 is eating...
Philosopher 3 finished eating.
Philosopher 2 is eating...
Philosopher 2 finished eating.
Philosopher 1 is eating...
Philosopher 1 finished eating.

hp@LAPTOP-E3MDDK1P MSYS ~
$
```